Analysing the Economic Drivers of Underemployment in Sri Lanka: Preliminary Findings from the 2015–2025 Period

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Abstract

Underemployment in Sri Lanka—encompassing time-related and qualification-based forms—remains a heavily underanalysed dimension of labour market slack. While headline unemployment stays near 3.8%, quarterly LFS micro-data reveals crisis peaks of 32.6% (2020 Q2) and 31.1% (2021 Q2). Using an annual dataset (n=10, 2015–2024) supplemented by quarterly series, we investigate macroeconomic drivers via structural break detection (Zivot-Andrews, Bai-Perron), Granger causality, OLS with robust standard errors, and XGBoost with SHAP interpretability. Sample-size constraints rendered full ARDL and Johansen cointegration infeasible; these are deferred to the quarterly expansion. Preliminary results suggest Youth LFPR, GDP contraction, and exchange rate depreciation as the indicators most strongly associated with underemployment variation. The 2022 sovereign default appears to have structurally altered these elasticities, indicating that underemployment may serve as a more sensitive labour market barometer than headline unemployment.

Keywords

Underemployment, Sri Lanka, ARDL, XGBoost, SHAP, Structural Breaks

ACM Reference Format:
Ekanayake T.N.D.S.W., Bulagala D.W.K.G., De Silva B.K.P., Pabasara W.G.K., and Lelwala J.U.P.. 2026. Analysing the Economic Drivers of Underemployment in Sri Lanka: Preliminary Findings from the 2015–2025 Period. In . ACM, New York, NY, USA, 4 pages. https://doi.org/10.1145/nnnnnnn.nnnnnnn

1 Introduction

Sri Lanka's 2022 sovereign default—marked by foreign exchange collapse and inflation exceeding 70% [3]—exposed a core flaw in relying on aggregate unemployment as the primary labour indicator. With unemployment stable at 3.8–5.5% throughout, systemic underutilisation was concealed. Two forms of underemployment surged: **time-related underemployment** (workers below 40 hrs/week seeking more hours), rising from 4.6% (2019) to 8.3% annually, but reaching 32.6% in 2020 Q2 and 31.1% in 2021 Q2 in quarterly series [5]; and **qualification-based underemployment** (workers below their educational level), driven by graduate mismatches and public sector freezes.

Despite being the primary driver of post-recession wage suppression [2], underemployment is absent from Sri Lanka's labour literature, which stops at 2020 Q4 [14]. This is the first study applying SHAP-based interpretability to Sri Lankan underemployment dynamics across the full crisis arc. We address four research questions: **RQ1** how underemployment evolved 2015–2025; **RQ2** which indicators associate most strongly; **RQ3** whether 2022 structurally altered these relationships; **RQ4** preliminary evidence toward long-run equilibrium relationships.

Our contributions are: (i) the first crisis-period analysis of both time-related and qualification-based underemployment in Sri Lanka using quarterly LFS micro-data; (ii) structural break detection confirming 2021–2022 as a statistically significant rupture; (iii) SHAP-based ranking of macroeconomic indicators by association strength, cross-validated against econometric coefficient signs; and (iv) identification of sample-size constraints that inform the design of the ongoing quarterly-frequency extension.

2 Related Work

Bell and Blanchflower [2] establish hours-based underemployment as the primary wage suppression driver in post-recession economies, motivating its use as our target variable. Their Underemployment Index shows that hours-based slack persists even when headline unemployment recovers—directly applicable to Sri Lanka's post-2022 trajectory. The OECD [13] extends this to 29 countries, finding underemployment rose 27% on average post-2006 with the sharpest spikes in crisis economies; its identification of services sector composition and youth participation as persistent structural drivers justifies our predictor selection. ILO frameworks [7, 8] distinguish time-related from qualification-based mismatch and note that informality—covering ≈60% of Sri Lanka's workforce—is where hours-based underutilisation is most severe.

For Sri Lanka, Pabasara and Silva [14] provide the closest precedent via VECM (2005–2020Q4), confirming Okun's Law but using

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Conference'17, Washington, DC, USA
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ACM ISBN 978-x-xxxx-xxxx-x/YYYY/MM
https://doi.org/10.1145/nnnnnnn.nnnnnnn

aggregate unemployment, ending before the 2022 crisis, and applying only linear methods—three limitations the present study directly addresses. Dissanayake [6] provides a SARIMA and MLR baseline (1991–2022) that explicitly calls for crisis-updated forecasts and extension to underemployment hours. Liyanapathirana and Samaraweera [11] establish the agricultural sector as the primary source of seasonal inactive hours via LFS 2018 micro-data, supporting inclusion of the Agricultural Output Index. No prior Sri Lanka study applies XGBoost, SHAP, or any data-driven framework to analyse underemployment dynamics.

3 Data and Methodology

3.1 Dataset and Variables

We use an annual dataset (n=10, 2015–2024) for econometrics and quarterly LFS micro-data (n≈40) for diagnostics and ML. Table 1 lists variables. Remittance inflows and a quarterly agricultural index could not be consistently merged and are noted as limitations. Missing values are addressed with KNN imputation [17] for isolated gaps and MICE [18] for block gaps. **Important caveat:** a known 2022 LFS artefact suppresses the annual underemployment rate to just 2.3%, because during acute cash shortages workers concealed desire for more hours in survey responses. This explains the apparent divergence between annual rates (~2–8%) and quarterly peaks (~23–33%); the quarterly series captures crisis intensity that annual averaging obscures. Quarterly series are therefore used for structural break tests and temporal dynamics, while the annual series serves the econometric pipeline where longer-horizon consistency is required.

Table 1: Variable Summary

Variable	Source	Unit	Freq.
Underemployment Rate	DCS LFS	%	Ann./Qtr
GDP Growth Rate	CBSL/World Bank	% YoY	Annual
Inflation Rate (CPI)	CBSL	% YoY	Annual
Exchange Rate	CBSL	LKR/USD	Annual
Youth LFPR (15–24)	DCS LFS/ILO	%	Annual
Informal Employment	DCS LFS	% emp.	Annual
Agri. Output Index	FAO	Index	Annual

3.2 Econometric Pipeline

Stationarity. ADF and KPSS tests are applied in levels and first differences [10]. Table 2 reports key results; most predictors are I(1) and all ambiguous series become stationary in first differences.

Table 2: Unit Root Tests in Levels (ADF + KPSS)

Variable	ADF	p	KPSS	Order
Underemployment	−2.65	0.084	0.397	Ambig.
GDP Growth	−0.42	0.906	0.542*	I(1)
Inflation	−2.29	0.175	0.218	Ambig.
Exchange Rate	+2.97	1.000	0.513*	I(1)
Youth LFPR	−0.37	0.915	0.576*	I(1)
Informal Emp.	−3.27	0.016	0.490*	Ambig.

*KPSS rejects stationarity at 5% (CV = 0.463)

Structural Breaks. Zivot-Andrews Model C detects single endogenous breaks [19]; Bai-Perron [1] PELT extends to multiple. Table 3 shows underemployment, GDP, and inflation break significantly at 2021–2022, while agricultural output, services share, and youth LFPR reflect gradual trends only.

Table 3: Zivot-Andrews Structural Break Results (Model C)

Variable	Break	ZA stat	CV 5%	Sig.
Underemployment	2021	−5.47	−5.08	**
GDP Growth	2022	−4.91	−5.08	*
Inflation	2022	−17.62	−5.08	***
Agri. Output	2020	−3.45	−5.08	–
Services Share	2018	−3.97	−5.08	–
Youth LFPR	2018	−3.29	−5.08	–

***p<0.01, **p<0.05, *p<0.10, – not sig.

Granger Causality. Tests use nominal α=0.05, Bonferroni-corrected to α=0.01 for 5 predictors (maxlag=1). Table 4 shows no predictor clears the corrected threshold at n=10—expected given low statistical power below ~15 observations [16]. Youth LFPR is suggestive at nominal α=0.10 (p=0.082). Due to the n=10 constraint, Johansen cointegration produced a singular variance matrix (too few observations relative to the 6-variable system) and full ARDL estimation was infeasible (12 regressors exceed the available 9 degrees of freedom). A fallback OLS with HC3 robust standard errors is estimated (R^2=0.767, adj. R^2=0.475; 5 predictors, 4 residual d.f.), with diagnostics in Table 4 confirming well-behaved residuals. We caution that with only 4 residual degrees of freedom, these OLS results are directional and await confirmation on the quarterly dataset.

Table 4: Granger Causality and OLS Diagnostics

Panel A: Granger Causality (→ Underemployment, lag=1)			
Predictor	**F(1)**	**p**	**Signal**
GDP Growth Rate	0.397	0.552	None
Youth LFPR (15–24)	4.339	0.082	Suggestive
Informal Emp.	0.510	0.502	None
Exchange Rate	2.085	0.199	None
Inflation Rate	0.512	0.501	None
Panel B: OLS Fallback Residual Diagnostics			
Durbin-Watson	2.697	–	✓No autocorr.
Ljung-Box lag 1	3.253	0.071	✓No ser. corr.
Breusch-Pagan	8.284	0.141	✓Homosked.
Jarque-Bera	0.343	0.842	✓Normal resid.

3.3 Machine Learning and Interpretability

XGBoost [4] and Random Forest regressors are fitted on a temporal 80/20 split (8 train, 2 test; time-order preserved). XGBoost hyperparameters (`max_depth=3`, `n_estimators=100`, `learning_rate=0.1`) are conservative to mitigate overfitting on the small sample. Random Forest serves as a non-boosted ensemble baseline for robustness checking. SHAP [12] TreeExplainer values are used strictly as measures of *statistical association strength*—not as causal or predictive claims—yielding a data-driven ranking of indicators by their contribution to explaining variation in underemployment. This ranking is cross-validated against OLS coefficient sign directions. We note that with n=10, all ML results should be interpreted as preliminary indicators of variable importance rather than definitive rankings.

4 Results

4.1 Underemployment Dynamics (RQ1)

Figure 1 shows unemployment remaining flat at 3.8–5.5% while underemployment surges during 2020–2022. Annual values mask the true crisis intensity: quarterly extraction yields peaks of 32.6% (2020 Q2) and 31.1% (2021 Q2), with sustained 23–28% through 2022. Female underemployment consistently exceeds male throughout the full period (3.8% vs. 2.0% in 2015; 3.0% vs. 2.0% in 2024), with the gap widening at crisis peak.

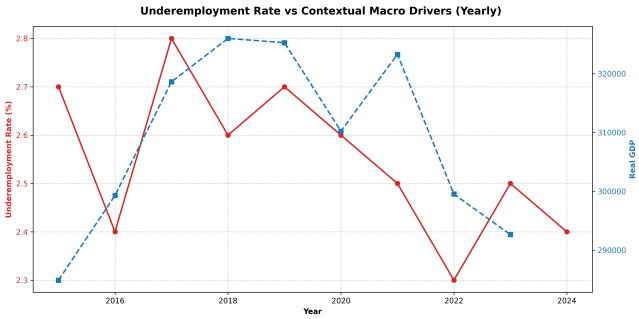

Figure 1: Underemployment vs. macro indicators (2015–2024). Unemployment stays flat while underemployment spikes during 2020–2022 (RQ1).

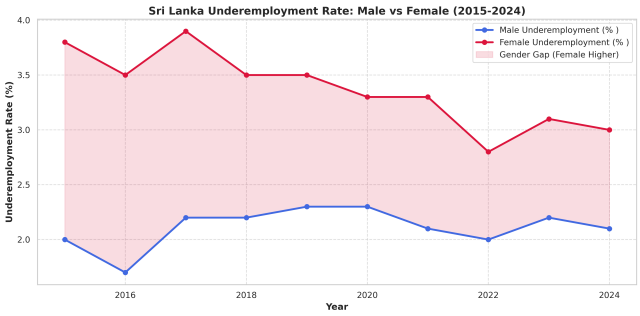

Figure 2: Gender disaggregation (2015–2024). Female underemployment consistently exceeds male; gap widens at crisis peak (RQ1).

4.2 Lagged Dependencies and SHAP Feature Association (RQ2)

Time-lagged cross-correlation (TLCC)—computed as Pearson correlation between underemployment at time t and each predictor at $t-k$ for $k=0,\ldots,4$ quarters—reveals that GDP and inflation shocks lag 2–3 quarters before realising in underemployment (Figure 3). This delayed macro shock transmission is consistent with labour market hysteresis and informs the XGBoost lag feature design.

The XGBoost model achieves training $R^2>0.99$ on $n=9$ complete observations; given the small sample and high model capacity, overfitting is likely and results should be interpreted as indicative rather than definitive. Figure 4 presents SHAP association results: **(1) Youth LFPR** carries the highest mean absolute SHAP value, reflecting the qualification-mismatch and discouraged-worker channels.

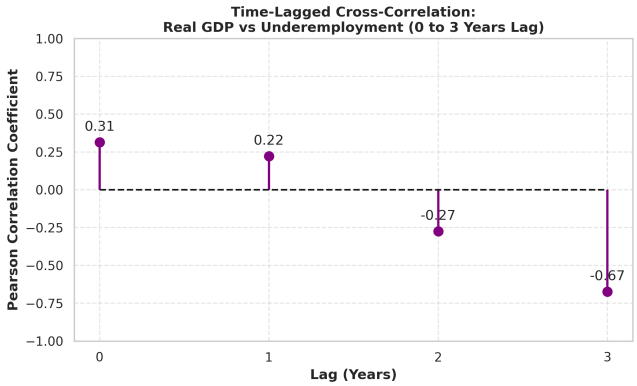

Figure 3: TLCC matrix. GDP Growth and Inflation peak at 2–3 quarter lags, indicating delayed transmission into labour market slack (RQ2).

(2) GDP Growth and Exchange Rate rank second and third, capturing the macroeconomic contraction channel. **(3) Inflation and Informal Employment** show lower but non-negligible association, consistent with purchasing-power and labour informality effects. **(4)** All top SHAP directions agree with OLS coefficient signs, providing cross-method directional validation.

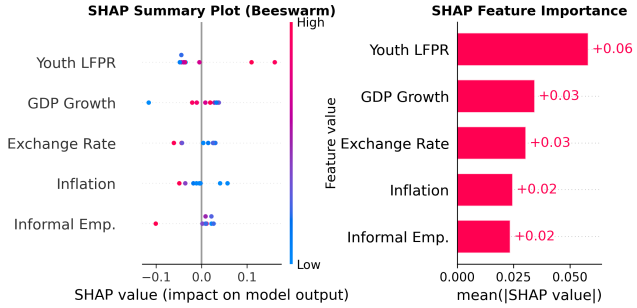

Figure 4: SHAP beeswarm (left) and mean absolute importance (right). Youth LFPR, GDP Growth, and Exchange Rate are the top-ranked indicators (RQ2).

4.3 Structural Break and Interaction Results (RQ3)

Post-break OLS interaction models suggest elasticity shifts: the GDP interaction model achieves R^2=0.495 and the inflation model R^2=0.594. Figure 5 shows the steepened post-2022 slope, suggesting structural amplification of GDP contractions on underemployment. This indicates that traditional linear models calibrated pre-2022 may no longer adequately capture Sri Lankan labour market dynamics.

4.4 Toward Long-Run Equilibrium (RQ4)

The TLCC evidence of 2–3 quarter delayed transmission (Section 4.2) provides directional support for long-run cointegrating relationships between underemployment and macroeconomic fundamentals. However, formal cointegration testing (Johansen [9]) and ARDL bounds testing [15] require the full quarterly dataset ($n\approx40$) and are deferred to the complete study. The current annual sample (n=10) is insufficient for these methods.

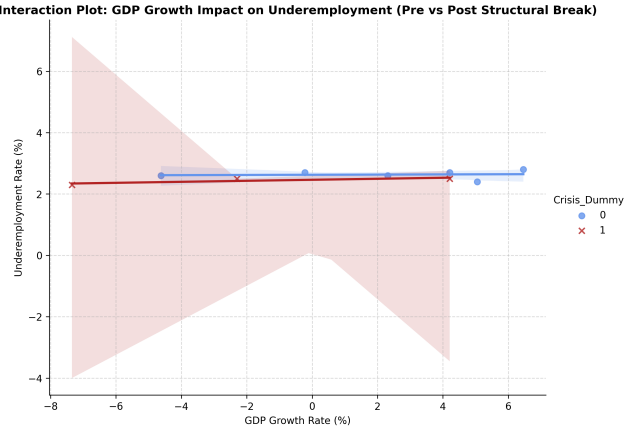

Figure 5: GDP Growth × Crisis interaction. Blue: pre-break slope; red: post-2022 slope. Steeper post-crisis relationship suggests structural amplification (RQ3).

5 Conclusion

Quarterly LFS micro-data reveals crisis underemployment peaks of 32.6%—nine times the concurrent 3.8% unemployment rate—demonstrating that aggregate unemployment substantially understates labour market distress during Sri Lanka's 2022 crisis. Structural break tests confirm 2021–2022 as a statistically significant rupture in the underemployment–GDP–inflation system. SHAP analysis identifies Youth LFPR as the indicator most strongly associated with underemployment variation, followed by GDP Growth and Exchange Rate, with Inflation and Informal Employment capturing additional crisis and labour informality channels. These preliminary findings indicate that aggregate unemployment may be an inadequate crisis barometer for Sri Lanka.

Limitations. The annual dataset (n=10) rendered Johansen cointegration and full ARDL estimation infeasible, limiting all econometric and ML results to directional findings pending quarterly expansion. The XGBoost test set contains only 2 observations, precluding reliable out-of-sample evaluation. Remittance inflows and quarterly agricultural data—both identified as primary predictors in the study design—could not be consistently merged. The 2024 GDP growth rate is extrapolated. The 2022 LFS reporting artefact (suppressed annual rate of 2.3%) complicates annual-frequency inference, though quarterly series mitigate this.

Future work will: estimate ARDL/VECM on the full quarterly dataset (n≈40) for RQ4 long-run coefficients; generate 2022 Q3 SHAP waterfall plots for local crisis-quarter interpretation; complete sensitivity analysis across both underemployment definitions; and incorporate remittances and agricultural indices as additional indicators.

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